

SASEBO, Japan

WMO: 478120

Lat: 33.1597N

Lon: 129.7275E

Elev: 6

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	1.5	-8.6	1.8	4.6	-7.3	2.0	4.7	9.5	6.7	8.9	6.7	2.9	80	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	33.2	25.5	32.1	25.5	31.2	25.2	26.8	30.6	26.4	30.0	26.0	29.5	4.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.2	28.8	25.5	20.7	28.6	25.0	20.1	28.4	84.4	30.7	82.5	30.1	80.8	29.6	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.9	6.8	-1.5	35.3	1.0	1.1	-2.2	36.1	-2.8	36.7	-3.3	37.3	-4.1	38.1		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-3.3	27.6	0.9	0.9	-3.9	28.2	-4.5	28.7	-5.0	29.2	-5.7	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.8	7.8	10.8	15.3	19.7	23.2	27.2	28.5	25.2	20.2	14.4	9.1
	DBStd	7.83	2.67	3.07	2.98	2.80	2.03	1.84	2.06	1.73	2.34	2.64	3.16	2.98
	HDD10.0	261	104	74	26	1	0	0	0	0	0	3	53	
	HDD18.3	1409	357	294	233	95	9	0	0	0	12	123	286	
	CDD10.0	2968	5	14	50	161	302	396	534	456	316	134	26	
	CDD18.3	1074	0	0	0	5	53	146	276	315	206	69	4	0
	CDH23.3	9170	0	0	0	9	167	693	2810	3662	1586	240	3	0
	CDH26.7	2914	0	0	0	0	12	90	918	1461	407	27	0	0
Wind	WSAvg	2.9	3.0	3.0	3.2	3.0	2.7	2.6	2.9	2.8	2.8	2.7	2.6	3.0
Precipitation	PrecAvg	2022	70	87	130	167	186	331	325	241	197	98	112	78
	PrecMax	2812	189	171	259	338	327	545	552	615	548	228	225	184
	PrecMin	1080	17	24	50	70	65	35	41	30	67	4	13	5
	PrecStd	402	39	42	48	74	73	134	154	143	111	66	58	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	18.6	20.8	24.6	28.2	30.1	34.4	35.2	32.7	29.1	23.8	18.4
		MCWB	13.5	14.6	14.9	17.3	19.0	22.3	25.8	25.9	24.6	21.8	18.1	14.5
	2%	DB	14.1	16.6	19.1	22.8	26.3	28.7	32.7	33.8	31.3	27.2	22.1	16.8
		MCWB	10.7	13.0	13.9	16.5	18.4	22.1	25.6	25.5	24.2	20.8	17.1	12.5
	5%	DB	12.7	14.6	17.5	21.6	25.1	27.7	31.5	32.7	30.1	25.7	20.7	15.4
		MCWB	9.1	11.1	13.3	15.9	18.3	22.2	25.6	25.6	23.7	19.7	16.4	11.4
	10%	DB	11.4	13.0	16.0	20.3	24.0	26.7	30.6	31.8	29.0	24.6	19.5	14.1
		MCWB	7.9	9.4	11.9	15.2	18.0	22.2	25.4	25.5	23.3	18.9	15.5	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.9	17.1	20.4	22.8	25.5	27.6	27.5	26.4	24.4	20.1	16.3
		MCDB	15.8	17.3	18.8	22.2	24.6	27.8	30.8	31.6	29.5	27.1	21.6	17.3
	2%	WB	11.8	13.9	15.4	18.7	21.4	24.9	26.8	27.0	25.7	22.7	18.7	13.7
		MCDB	13.3	15.9	17.6	21.0	23.4	26.9	30.1	31.1	29.0	25.7	20.7	15.6
	5%	WB	9.7	11.8	14.0	17.6	20.4	24.2	26.4	26.6	25.1	21.3	17.4	12.1
		MCDB	11.9	14.0	17.0	20.2	22.8	26.2	29.6	30.5	28.6	24.5	20.1	14.6
	10%	WB	8.2	9.8	12.6	16.3	19.4	23.4	26.0	26.2	24.4	19.9	16.0	10.8
		MCDB	11.0	12.5	15.5	19.3	22.5	25.5	29.2	30.0	27.9	23.3	19.0	13.6
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.9	7.3	7.5	7.5	5.9	5.2	5.9	6.2	7.0	7.0	6.6
		MCDBR	8.2	8.8	9.2	9.4	9.4	7.6	6.7	7.2	7.3	8.2	8.0	8.1
		MCWBR	6.0	6.4	6.3	5.3	4.6	3.5	2.5	2.7	3.2	4.1	4.9	5.7
	5% WB	MCDBR	7.3	7.4	7.8	6.8	6.3	4.5	5.6	6.1	5.9	6.3	6.6	7.1
		MCWBR	6.4	6.8	6.7	5.2	4.5	2.9	2.5	2.7	3.0	4.1	5.1	6.0
Clear-Sky Solar Irradiance	taub	0.451	0.492	0.554	0.574	0.582	0.597	0.577	0.548	0.519	0.474	0.445	0.440	
	taud	2.004	1.933	1.793	1.800	1.839	1.891	1.983	2.067	2.087	2.105	2.105	2.043	
	Ebn at Noon	726	741	729	738	738	724	736	751	753	752	731	711	
	Edh at Noon	142	168	208	216	209	198	180	163	154	140	127	129	
All-Sky Solar Radiation	RadAvg	2.14	2.90	4.02	4.85	5.21	4.45	5.12	5.33	4.27	3.57	2.57	1.99	
	RadStd	0.21	0.30	0.27	0.41	0.61	0.60	0.74	0.50	0.43	0.35	0.26	0.15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page